

Problem file:

Problem 1: Entropy Calculation

symbols with probabilities:

- A: 0.5

- B: 0.25

- C: 0.25

Compute entropy.

Solution:

$H = -(0.5 \log_2 0.5 + 0.25 \log_2 0.25 + 0.25 \log_2 0.25) = -(0.5 * -1 + 0.25 * -2 + 0.25 * -2) = 1.5$
bits

Problem 2: Data vs Information

If English text has entropy = 1.5 bits/char, how efficient is ASCII?

Efficiency = $1.5 / 8 = 18.75\%$

Problem 3: Base64 Overhead

3 bytes to 4 bytes

$(4 - 3) / 3 = 33\%$